Ezra Ornstein

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Skills

Tools: Altium, LTspice, Oscilloscope, Digital Multimeter, Soldering, Harnessing, SOLIDWORKS, AutoCAD
Languages and Protocols: C, C++, Git, PWM, DShot, I2C, SPI, UART, CAN, USB PD

Experience

Electrical Engineering Co-op – Waterloo Aerial Robotics Group May 2026 – Aug 2026
University of Waterloo

- Eliminated firmware team's Li-ion dependency for testing by designing an ESP32-based USB-C PD sink PCB using Altium and LTspice to design and simulate the board and I2C for onboard communication
- Utilized Altium and LTspice to design and simulate a high-efficiency buck converter PCB to step UAV battery down to 12V to power drone peripherals, considering EMI and parasitic inductance and capacitance
- Assisted electrical team members with schematic and layout reviews, board assembly and testing, and component requisitions, ensuring boards were designed and assembled correctly and efficiently

Engineering Projects

USB-C PD Sink With PPS Control — *Altium, LTspice, Oscilloscope, DMM, Soldering, C* Jun 2026 – Jul 2026

- Designed, assembled and tested a USB-C PD sink board supporting PPS negotiation via an ESP32-C3-MINI MCU communicating with a USB PDIC over I2C as well as an OLED screen and voltage and current sensors
- Implemented four configurable 5V-20V power outputs at up to 4A and a fixed 3V3 header at up to 1A to support diverse downstream loads, allowing the firmware team to test any system they need with one board
- Programmed ESP32 firmware and web server connected over Wi-Fi to phones and laptops for in-depth readout

Custom FPV Flight Controller — *Altium, LTspice, Soldering, Digital Multimeter, Oscilloscope* Ongoing

- Designed a 6-layer 30.5×30.5mm flight controller around an STM32 MCU with onboard barometer and IMU, verifying full component compatibility for Betaflight integration
- Integrated multiple SPI, I2C and UART connections, a USB-C port with differential routing to power the board and flash the MCU, and added connectors and pads for modular drone assembly
- Integrated multistage power system with buck converters from the battery to 5V and 9V, an Or-ing diode to support USB and 5V simultaneously, and an 3V3 LDO to power the MCU and peripherals

VTOL UAV Wiring Architecture — *Altium, SOLIDWORKS, Harnessing, Soldering, DMM* Ongoing

- Designed the complete power and signal harnessing for a 2.5m wingspan, 5-motor VTOL UAV, wiring 4 lift motors and 1 pusher, dual 6S3P battery packs and a Pixhawk flight controller
- Engineered per-wing harness architecture including dual XT60 power connections, 2 ESCs, a custom servo hub and 5 servo connectors, and a CAN bus network integrating barometer, gyroscope, and GPS sensors

High Efficiency 48V - 12V Buck Converter — *Altium, LTspice, Oscilloscope* May 2026 – Jun 2026

- Designed, simulated, assembled, and validated a 30.5 × 30.5 mm high-efficiency buck converter PCB to step 12S UAV battery voltage (36–50.4 V) down to 12 V, achieving 95% peak efficiency
- Utilized Oscilloscope and Digital Multimeter to perform full testing from source to sink, probing input and output filters as well as switching node and testing during both steady state and transient events

Electric Go-Kart Conversion — *Harnessing, Soldering, Digital Multimeter* Aug 2026 – Sep 2026

- Converted a gas-powered (lawnmower-engine) go-kart to full electric drive, redesigning the powertrain and control systems and acheiving 1800W of power at 48V with up to an hour of runtime
- Installed and harnessed new 48V 1800W BLDC motor, motor controller, 12S3P battery, ignition key, forward/reverse and 3-speed switch controls for variable drive modes

Education

University of Waterloo Waterloo, ON
Bachelor of Applied Science in Mechatronics Engineering *Sep 2025 - Present*